FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Neonatal Resuscitation

Clinical skill | Week 5 · Bleeding, cancer & fertility | Dr Nassimu

About 1 in 10 newborns needs help to start breathing and about 3-6 in 100 need bag-and-mask ventilation. Prepare before every birth, dry and stimulate, and if the baby is not breathing, start effective bag-and-mask ventilation within the Golden Minute; ventilation (not suction, not drugs) is the single most important intervention.

Must know before the ward round

1. Golden Minute - a baby who is not breathing well after drying and stimulation must be receiving bag-and-mask ventilation within 60 seconds of birth.
2. Prepare before every birth - helper identified, warm area, towels and hat, bag tested, masks size 1 and 0, suction, stethoscope, clock.
3. Effective ventilation = gentle chest rise at 40 breaths per minute (acceptable 30-60) using a size 1 mask for term and size 0 for preterm babies; start with room air for term babies.
4. If the chest does not rise - improve ventilation - reapply the mask, reposition the head (slightly extended), clear the airway, open the mouth, squeeze harder.
5. Heart rate is the best indicator of response - normal is 100 or above; below 100 means continue ventilating and call for help; below 60 despite 30 seconds of effective ventilation means chest compressions (3 - 1) where advanced care is available.
6. Gasping is NOT breathing - ventilate.
7. Suction only if the airway is blocked (secretions or meconium and the baby is not crying); no routine deep suction and no routine tracheal suction for meconium.
8. Delay cord clamping 1-3 minutes in babies who breathe; if the baby needs ventilation and the setup does not allow ventilating on the cord, clamp and cut and move to the resuscitation area.
9. Adrenaline is 1:10,000 (0.1 mg/mL) at 0.1-0.3 mL/kg IV (0.01-0.03 mg/kg); to make it, dilute 1 mL of 1:1,000 in 9 mL saline.
10. After resuscitation - check temperature (36.5-37.5), glucose (treat below 2.6 mmol/L with 10% dextrose 2 mL/kg), breathing and tone every 15 min for 2 h; SCBU if lethargic, poor suck or seizures.
11. APGAR is recorded at 1 and 5 minutes but is NEVER used to decide whether to start resuscitation.
12. Stop if no heart rate after 10 minutes of effective ventilation (WHO); consider stopping if heart rate stays below 60 with no breathing after 20 minutes.

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1. Purpose and Indications

Neonatal resuscitation is the set of actions that help a newborn establish breathing and circulation at birth. Most babies breathe on their own. About **10%** need some help (drying, stimulation, airway clearance) and roughly **3-6%** need **bag-and-mask ventilation**; fewer than 1% need chest compressions or drugs. That means **ventilation fixes almost every baby who needs help**.

Helping Babies Breathe (HBB), developed by the American Academy of Pediatrics with WHO and partners and taught widely in Uganda through the Ministry of Health, is a simple, evidence-based programme for low-resource settings built around the **Golden Minute**: within 60 seconds of birth, a baby who is not breathing must be dried, stimulated and ventilated. HBB is now folded into the WHO **Essential Newborn Care 1** course (2024 provider guide).

> CLINICAL PEARL
> The one thing never to forget: **ventilation is the key**. Every second a non-breathing baby goes without ventilation increases the risk of death or brain injury. Do not waste the Golden Minute on repeated suction, slapping, or looking for oxygen.

Definitions

Term	Meaning
Golden Minute	The first 60 seconds after birth, within which ventilation should begin for a baby not breathing well
Primary apnoea	Early response to hypoxia: breathing stops, heart rate falls, but stimulation can restart breathing
Secondary (terminal) apnoea	Prolonged hypoxia: gasping then apnoea; the baby **will not respond to stimulation** and needs ventilation; untreated leads to death
Birth asphyxia (WHO clinical)	Failure to initiate and sustain breathing at birth
Hypoxic-ischaemic encephalopathy (HIE)	Brain injury after perinatal asphyxia; graded mild/moderate/severe (Sarnat)
Vigorous baby	Strong respiratory effort (crying/breathing well), good tone, heart rate 100 or above
Effective ventilation	Ventilation that produces **visible, gentle chest rise** with each breath
Advanced resuscitation	Chest compressions, oxygen titration, intubation, umbilical venous access, adrenaline and volume: in hospitals/SCBU

Levels of resuscitation

1. **Basic (every birth attendant, every facility)**: drying, warmth, airway clearance if needed, stimulation, **bag-and-mask ventilation**, heart rate assessment (HBB).
2. **Advanced (hospital, SCBU/NICU)**: oxygen and pulse oximetry, chest compressions, intubation, adrenaline, volume expansion (NRP / Uganda newborn guidelines).

Why it matters in Uganda

- **Birth asphyxia** (intrapartum-related events) is one of the three leading causes of newborn death, alongside prematurity and infection, and a major cause of fresh stillbirths and cerebral palsy. Many of these babies could survive intact with **a trained person, a working bag and mask, and no delay**.
- Uganda's neonatal mortality rate is about **22 per 1,000 live births** (UDHS 2022), meaning tens of thousands of newborn deaths a year; intrapartum-related complications (asphyxia) account for roughly a quarter to a third of them Verify *share of neonatal deaths due to asphyxia; no Uganda-specific source cited.*

- **HBB** was introduced in Uganda about a decade ago and is part of the MoH **Essential Newborn Care** training; many facilities have HBB kits (NeoNatalie simulator, bag, masks, penguin suction).
- **Realities**: at many HC IIIs and HC IVs there is one midwife conducting the delivery and resuscitating at the same time; bags and masks may be missing, broken or not cleaned; there may be no oxygen or pulse oximeter. **Daily equipment checks** and **practice on the simulator** save lives.
- **Stillbirth misclassification**: babies born apparently lifeless but not macerated may be *fresh* and may respond to ventilation. **Ventilate any baby who is not breathing unless macerated**.
- Referral level: **Mengo, Mulago/Kawempe, regional referral hospitals** have special care baby units (SCBU) with oxygen, CPAP (bubble CPAP is increasingly available), phototherapy and neonatal staff. Therapeutic hypothermia is not routinely available in most Ugandan units.

> **UGANDA CONTEXT**
> The Ugandan standard for basic newborn resuscitation at all levels follows **Helping Babies Breathe** and WHO basic newborn resuscitation guidance. Advanced steps (chest compressions, adrenaline, intubation) are done in hospitals with trained staff. Know HBB perfectly; know the advanced steps well enough to assist and to explain.

Indications: who needs what

Baby	Action
Crying/breathing well, good tone	Routine care: dry, skin-to-skin, delay cord clamping, breastfeed, monitor
Not crying after drying	Keep warm, clear airway if needed, stimulate (rub back 2-3 times)
Not breathing or gasping after stimulation	**Ventilate within the Golden Minute**
Heart rate below 100 after ventilation	Continue ventilation, improve it, call for help
Heart rate below 60 after 30 s of effective ventilation (chest rising)	Advanced: chest compressions with ventilation (100% oxygen if available), then adrenaline if still below 60

2. Contraindications and Precautions

When not to start

- **Do not start** for a baby with **obvious signs of death** (macerated stillbirth, severe decomposition) or **lethal anomalies** (e.g. anencephaly), or extremely preterm below the local threshold of viability (discuss with a senior; in Uganda many units use about 28 weeks or 1 kg as a practical threshold) Verify *no national written viability threshold cited*.
- A fresh (non-macerated) baby who looks lifeless is **not** a reason to withhold: ventilate.

When to stop

- **Stop** (WHO 2012): **no detectable heart rate after 10 minutes of effective ventilation**; **heart rate below 60 and no spontaneous breathing after 20 minutes** of resuscitation. Always involve a senior and support the family.

Precautions

- **Standard precautions** for every birth: gloves, hand hygiene; the baby of an HIV-positive mother is resuscitated exactly the same way.
- **Do not suction deeply or routinely**: vagal bradycardia, laryngospasm, mucosal trauma.
- **Do not over-ventilate**: big squeezes or rates over 60/min cause lung injury and pneumothorax.

- **Avoid hyperoxia** where oxygen is used: titrate to pre-ductal SpO_2 targets.
- **Avoid hyperthermia** after resuscitation: it worsens brain injury.
- **Congenital diaphragmatic hernia** (scaphoid abdomen, known on scan): avoid prolonged bag-mask ventilation, intubate.
- **Never use 1:1,000 adrenaline undiluted** in a neonate.
- **Do not transfer** a baby who is still being bag-ventilated without a health worker continuing ventilation throughout.

3. Relevant Anatomy and Normal Values

Transition at birth

- In utero, the lungs are **filled with fluid**, pulmonary vascular resistance is high, and blood bypasses the lungs through the **foramen ovale and ductus arteriosus**. The placenta does gas exchange.
- At birth, the first breaths (triggered by cold, touch, CO_2 rise and cord clamping) **push lung fluid into the interstitium**, open the alveoli, raise oxygen levels, and cause **pulmonary vasodilation**. Pulmonary blood flow increases, left atrial pressure rises, the **foramen ovale closes**, and the **ductus begins to constrict**.
- If the baby **does not breathe**, the lungs stay fluid-filled, pulmonary vascular resistance stays high, hypoxia worsens, and the heart slows (**bradycardia is a sign of hypoxia** in newborns, not a primary heart problem).

Primary vs secondary apnoea

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Hypoxia begins
    |
Rapid breathing -> PRIMARY APNOEA (HR falls; responds to stimulation)
    |
Irregular GASPING (HR falls further, BP falls)
    |
SECONDARY/TERMINAL APNOEA (will NOT respond to stimulation;
    |                      needs ventilation; drugs rarely help)
Death / brain injury if no ventilation
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At birth you **cannot tell** whether an apnoeic baby is in primary or secondary apnoea. So if a baby does not breathe after brief drying and stimulation, **assume secondary apnoea and ventilate**.

Why the heart rate improves with ventilation

The newborn myocardium is hypoxic and slows down. **Oxygenating the lungs** delivers oxygenated blood to the coronary arteries, and the heart rate rises. A rising heart rate is therefore the **best sign that your ventilation is working**. This also explains why **chest compressions are useless without effective ventilation**, and why adrenaline is rarely needed.

Why keep the baby warm

Newborns lose heat rapidly (wet skin: evaporation; large surface area). Cold stress increases **oxygen consumption**, causes **hypoglycaemia and acidosis**, and worsens outcomes, especially in preterm babies. Every 1°C below 36.5°C increases mortality in preterm infants. Drying is both warming and stimulating.

Delayed cord clamping

Waiting **1-3 minutes** in breathing babies transfers placental blood (improves iron stores and blood pressure, and in preterm babies reduces intraventricular haemorrhage and need for transfusion). HBB 2nd edition allows ventilation with the cord intact where equipment allows, but **ventilation must not be delayed** to wait for

clamping.

Anatomy for the procedure

- **Airway position**: head **slightly extended** ("sniffing"); the newborn occiput is large, so a small roll under the shoulders helps. Over-extension or flexion both kink the soft airway.
- **Mask fit**: a round, soft-rimmed mask covering the **chin, mouth and nose** but not the eyes. Size 1 for term, size 0 for preterm/small.
- **Heart rate**: listen over the **left chest** (most reliable), or feel the pulsation at the **base of the umbilical cord** (may underestimate).
- **Compressions**: **lower third of the sternum**, depth about **one-third of the AP diameter** of the chest.
- **Umbilical vein**: the single thin-walled large vessel (two thick-walled arteries) used for emergency venous access (UVC).

Normal values

Item	Normal value
Respiratory rate	**40–60/min**
Heart rate	**100–160/min** at rest; **≥100** is the resuscitation threshold for "normal"
Temperature	**36.5–37.5°C**
Pre-ductal SpO_2 (right hand)	About 60–65% at 1 min, 65–70% at 2 min, 70–75% at 3 min, 75–80% at 4 min, 80–85% at 5 min, **85–95% at 10 min**; ≥90% once stable
Blood glucose	Treat if **<2.6 mmol/L** (Uganda/WHO practice; NICE uses 2.0 mmol/L in the first 48 h)
Hb at birth	About **14–20 g/dL**
Capillary refill	**<3 s**
Birth weight	**≥2,500 g** (below = LBW)
Room temperature for birth	About **25–28°C**
Self-inflating neonatal bag volume	About **240–320 mL**

4. Equipment and Preparation

The HBB kit and newborn corner (check before every birth)

Item	Check
Warm room/area (no drafts), resuscitation surface (flat, firm, warm)	Room about 25-28°C
Two or three clean, dry towels/cloths and a **hat**	One to dry, one to cover
Self-inflating bag (neonatal, about 240-320 mL)	**Test**: squeeze with the mask sealed against your palm; you should feel pressure and the pop-off valve should open; the bag re-inflates
Masks: size 1 (term) and size 0 (preterm/small)	Round, soft rim; fits over **chin, mouth and nose** but not the eyes
Suction device (penguin bulb or mucus extractor)	Clean, working
Clock/timer with seconds	To mark time of birth and the Golden Minute

Item	Check
Stethoscope	For heart rate
Cord clamps/ties, sterile scissors, gloves	
Advanced (hospital): oxygen source, blender, pulse oximeter, laryngoscope (blades 0 and 00), endotracheal tubes (2.5-3.5 mm), umbilical catheter, adrenaline 1:10,000, normal saline, 10% dextrose, radiant warmer, plastic bag for preterms	

> **NOTE**
> After each use, **decontaminate**: disassemble the bag and mask, wash with soap and water, disinfect (e.g. high-level disinfection by boiling or 0.5% chlorine per protocol), rinse, dry and reassemble, then **test**. A dirty or wrongly reassembled bag is useless in an emergency.

Step 0: before birth (the "first 5 minutes" start before the baby is born)

1. **Is resuscitation likely?** Preterm, thick meconium, abnormal FHR, prolonged/obstructed labour, breech, twins, maternal fever, pethidine in the last few hours, APH. If yes, **a second skilled person is at the birth**.
2. **Identify a helper** and review risk factors (preterm, meconium, prolonged labour, fetal distress, breech, twins, maternal fever, pre-eclampsia, sedation).
3. **Prepare the area**: warm, clean, well lit.
4. **Wash hands**; lay out towels, hat, bag, masks (size 1 and 0), suction, stethoscope, clock, cord clamps and scissors.
5. **Equipment check** (takes 1 minute): **bag tested** against your palm, masks, suction bulb, stethoscope, clock.

Focused history: what to find out before the baby is born, and why

Ask these (from the mother, the midwife and the partograph), ideally **before** the birth; each answer changes what you prepare or do afterwards.

Question	Why it matters
Gestational age	Preterm: mask size 0, plastic wrap/hat, lower oxygen, CPAP, SCBU; more likely to need ventilation
FHR pattern and liquor (late decelerations, bradycardia, thick meconium)	Predict asphyxia; skilled helper present before birth; possible HIE and meconium aspiration
Length of labour and second stage; obstructed labour; instrumental delivery	Asphyxia, birth trauma (subgaleal haemorrhage after vacuum)
Maternal fever ≥38°C, ROM >18 h, foul liquor, chorioamnionitis	Neonatal sepsis risk: antibiotics after resuscitation
Maternal drugs: pethidine/morphine within about 4 h, $MgSO_4$, general anaesthesia	Respiratory depression and hypotonia: **ventilate** (naloxone is not routinely recommended)
APH, abruption, cord accident, fetomaternal bleed	Neonatal hypovolaemia: pale baby, poor perfusion; may need volume 10 mL/kg
Maternal diabetes	Hypoglycaemia, macrosomia, shoulder dystocia (brachial plexus injury)
Multiple pregnancy	Extra team and equipment
Maternal HIV (viral load), syphilis, hepatitis B, Rh-negative	Prophylaxis and post-resuscitation care plan

Question	Why it matters
Known anomalies on scan (diaphragmatic hernia, gastroschisis, hydrops)	Special handling (e.g. avoid prolonged bag-mask in diaphragmatic hernia: intubate)

Consent, communication, positioning and exposure

- **Consent**: resuscitation is an emergency and proceeds without waiting for formal consent; tell the mother (and companion) before the birth that a helper is ready, and explain what is happening as you work if someone is free to do so.
- **Position**: baby on its back on a flat, firm, warm surface (or on the mother's abdomen for drying), head slightly extended; you stand at the head.
- **Exposure**: the whole baby must be visible to assess chest movement and tone, but **covered** to prevent heat loss; remove the wet cloth, cover with a dry one, hat on.
- **Hand hygiene and gloves** before touching the baby.

5. Procedure Step by Step

At birth, second by second

Time	Assessment	Action
0 s	Note the time of birth	Start the clock
0–30 s	Crying? Breathing? Tone?	**Dry thoroughly**, remove the wet cloth, keep warm
~30 s	Not crying: airway blocked?	Position (slight extension); **suction only if secretions or meconium block the airway**
~30–60 s	Breathing well after rubbing the back?	If not breathing or **gasping** → clamp/cut the cord, **ventilate by 60 s** at 40/min
60 s onwards	**Chest rising?**	If not → improve ventilation (mask, head, airway, mouth, squeeze harder)
After 1 min of ventilation	**Heart rate** (stethoscope, 6 s × 10)	≥100: continue until breathing. <100: continue, improve, call for help. <60 after 30 s of effective ventilation: compressions (where trained)
1 and 5 min	APGAR (recorded, not used to decide)	Document; repeat every 5 min until ≥7 (up to 20 min)

A. The first 30 seconds

1. **Note the time of birth.**
2. **Dry thoroughly**: head, face, body, arms, legs, back; **remove the wet cloth** and cover with a dry one. Drying also stimulates breathing.
3. **Is the baby crying?**
   - **Yes (crying)**: keep warm skin-to-skin with the mother, check breathing, **delay cord clamping 1-3 minutes**, start breastfeeding within an hour. **Routine care.**
   - **No (not crying)**: continue below.

B. If not crying

4. **Keep warm** (skin-to-skin or on a warm surface, covered, hat).
5. **Clear the airway if needed**: position the head **slightly extended** ("sniffing" position; a small roll under the shoulders helps). **Suction only if** there are secretions blocking the airway, or meconium-stained liquor and

the baby is not crying: **mouth first, then nose**, gently, not deep into the throat (vagal bradycardia, laryngospasm, trauma).

6. **Stimulate breathing**: rub the back firmly 2-3 times (or flick the soles). Do not slap, shake, hold upside down, or squeeze the chest.
7. **Is the baby breathing well?**
   - **Yes**: keep warm, monitor breathing and colour, clamp and cut the cord, keep with mother, breastfeed.
   - **No (not breathing, or gasping)**: **ventilate now**.

C. Ventilation (by 60 seconds)

8. **Clamp and cut the cord** (unless your setup allows ventilation with the cord intact) and move the baby to the ventilation area if needed.
9. **Position**: baby on its back, head slightly extended, you stand at the head.
10. **Mask**: size 1 for term, size 0 for small/preterm. Place it on the **chin first, then over the mouth and nose**; hold it with a **C-grip** (thumb and index finger on the rim), and use the remaining fingers to **lift the jaw** into the mask. It must not cover the eyes.
11. **Squeeze the bag** to give **40 breaths per minute** ("**Squeeze**-two-three, **Squeeze**-two-three"), using just enough pressure to produce a **gentle rise of the chest**.
12. **Use room air** for term babies (oxygen is not needed to start; do not delay ventilation to find oxygen).
13. **Watch the chest**: is it moving with each breath?

D. Improve ventilation if the chest is not moving

In this order:

1. **Reapply the mask** to get a better seal.
2. **Reposition the head** (slightly extended; not over- or under-extended).
3. **Clear the airway** (secretions).
4. **Open the mouth slightly**.
5. **Squeeze the bag harder**.
6. In hospital, an **alternative airway** (endotracheal tube or laryngeal mask) if still ineffective ("MR SOPA").

E. After 1 minute of effective ventilation

14. **Call for help** if the baby is still not breathing well.
15. **Check the heart rate** (helper listens with a stethoscope, or feel the cord pulse at its base): count for 6 seconds and multiply by 10.
    - **Normal (100 or above)**: continue ventilation until the baby breathes well on its own, then stop and monitor closely.
    - **Slow (below 100)**: **continue ventilation**, improve ventilation, and decide on **advanced care** (with oxygen, compressions if below 60).
16. Continue to ventilate until the baby **breathes spontaneously** and the heart rate stays normal. Then **monitor with the mother** (breathing, heart rate, colour, temperature, feeding).

> **EXAM FAVOURITE**
> The two outcome questions at the end of the Golden Minute: "**Is the chest moving?**" (if not, improve ventilation) and "**What is the heart rate?**" (100 or above: continue until breathing; below 100: continue, improve, call for help, advanced care).

F. Advanced resuscitation (hospital, trained staff)

Based on ILCOR/AHA NRP 2020 and WHO; follow the protocol of your SCBU.

Step	Detail
Oxygen	Term and 35 weeks or more: start in **21% (air)**; below 35 weeks: 21-30%; titrate to **pre-ductal SpO_2 (right hand)** targets: about 60-65% at 1 min, 65-70% at 2 min, 70-75% at 3 min, 75-80% at 4 min, 80-85% at 5 min, 85-95% at 10 min
Chest compressions	If **heart rate below 60 after at least 30 s of ventilation that moves the chest**. **Two-thumb technique** on the **lower third of the sternum**, depth about **one-third of the AP diameter of the chest**; ratio **3 compressions : 1 breath** (90 compressions + 30 breaths per minute: "one-and-two-and-three-and-breathe"); increase oxygen to **100%**; recheck HR after **60 s**
Airway	Intubation or laryngeal mask if bag-mask ventilation is ineffective or prolonged, or before compressions if skilled
Adrenaline (epinephrine) 1:10,000 (0.1 mg/mL)	**IV/intraosseous (umbilical venous catheter preferred): 0.01-0.03 mg/kg = 0.1-0.3 mL/kg**, repeat every 3-5 min if HR remains below 60. **Endotracheal (while IV access is obtained): 0.05-0.1 mg/kg = 0.5-1 mL/kg**
Volume	Normal saline **10 mL/kg** IV over 5-10 min if suspected blood loss (pale, poor perfusion, maternal APH, cord bleeding) and poor response
Glucose	After resuscitation, check and treat low glucose (section 8)

DRUG DOSING

Adrenaline in neonates is always the **1:10,000 strength (0.1 mg/mL)**. If only 1:1,000 (1 mg/mL) is available, **dilute 1 mL in 9 mL normal saline** first. A 3 kg baby gets 0.3-0.9 mL of 1:10,000 IV.

NOTE

The AHA/AAP neonatal resuscitation guidance was updated after the 2020 edition used here (2025 update); check it for any change to cord management, initial oxygen or adrenaline advice before teaching the advanced steps Verify *2025 AHA/AAP neonatal guideline changes not reviewed in these notes*.

G. The full algorithm at a glance (HBB-based, with advanced extension)

1. **Prepare** (helper, equipment checked, warm area).
2. **Birth**: note time. **Dry thoroughly**, remove wet cloth.
3. **Crying?** Yes: routine care (skin-to-skin, delay clamping, breastfeed).
4. **Not crying**: keep warm, **clear airway if needed**, **stimulate**.
5. **Breathing well?** Yes: routine care with close monitoring.
6. **Not breathing / gasping**: **cut cord, ventilate by 60 seconds** at 40 breaths/min with visible chest rise.
7. **Chest not moving**: improve ventilation (reapply mask, reposition head, clear airway, open mouth, squeeze harder).
8. **After 1 minute of ventilation, not breathing well**: **call for help**, check **heart rate**.
9. **HR 100 or above**: continue ventilating until breathing, then monitor.
10. **HR below 100**: continue ventilating, improve ventilation, advanced care (oxygen if available).
11. **HR below 60 after 30 s of effective ventilation** (hospital): **chest compressions 3:1** with 100% oxygen; consider intubation.
12. **HR still below 60 after 60 s of compressions + ventilation**: **adrenaline 0.01-0.03 mg/kg IV** (UVC), repeat every 3-5 min; consider volume (10 mL/kg normal saline).

13. **No heart rate after 10 minutes** of effective ventilation: consider stopping (senior decision); **HR below 60 without breathing after 20 minutes**: stop.

H. Special situations

- **Meconium-stained liquor**: if the baby is **crying/vigorous**: routine care. If **not vigorous**: clear the mouth and nose if secretions block the airway, and **start ventilation within the first minute**. **Routine tracheal suctioning or intubation is no longer recommended** (it delays ventilation without improving outcomes); it may be done only if the airway is obstructed and a skilled person is present.
- **Preterm baby**: size 0 mask, lower pressures, keep very warm (plastic bag/wrap up to the neck without drying the body in very preterm babies, hat, skin-to-skin/KMC once stable); consider CPAP early.
- **Baby of HIV-positive mother**: resuscitate exactly the same way (standard precautions); start infant ARV prophylaxis afterwards.
- **Baby born before arrival, cold and floppy**: warm while you ventilate.
- **Twins**: prepare two sets of equipment and two resuscitators.

I. Examination after resuscitation (Hutchison order, adapted for the newborn)

Once the baby is breathing with HR ≥100 (within the first hour, then repeated):

1. **Introduction and consent**: introduce yourself to the mother, explain, and gain her consent to examine the baby.
2. **Hand hygiene**.
3. **Position and exposure**: under a warmer or in a warm room, undress only as needed, keep the hat on; examine on the mother's lap or skin-to-skin where possible.
4. **General inspection from the end of the cot**: well or unwell; breathing pattern (grunting, indrawing); **colour** (central cyanosis, pallor, jaundice); **tone and posture** (flexed vs floppy); alertness; obvious anomalies or dysmorphism.
5. **Vital signs and growth**: RR for a full minute, HR, temperature, SpO_2 (if oximeter), capillary refill; weight; blood glucose.
6. **Head, face and neck**: caput, cephalhaematoma, **boggy swelling crossing sutures** (subgaleal), fontanelle tension, palate, facial symmetry.
7. **Respiratory**: inspection (recession, symmetry) → auscultation (air entry, symmetry); a scaphoid abdomen suggests diaphragmatic hernia.
8. **Circulation**: pulses, perfusion, heart sounds.
9. **Neurology (Sarnat)**: alertness, tone, suck, Moro, seizures.
10. **Limbs and birth trauma**: arm movement (Erb's palsy), clavicle.
11. **Complete the examination**: full top-to-toe check for anomalies; observe a breastfeed.
12. **Thank** the mother, re-dress and wrap the baby, **wash hands**, **summarise** with an assessment and plan.

6. Findings and Interpretation

During resuscitation

1. **Breathing**: crying, regular breathing, gasping (counts as NOT breathing), apnoea; rate (normal 40-60/min); grunting, chest indrawing.
2. **Heart rate**: stethoscope over the left chest (most reliable), or palpate the base of the cord (may underestimate); pulse oximeter/ECG in hospitals.
3. **Colour**: central cyanosis vs pink; pallor (anaemia/shock). Colour alone is unreliable in the first minutes.
4. **Tone and activity**: floppy vs flexed.

5. **Temperature**: 36.5-37.5°C is normal.
6. **After stabilisation**: full newborn exam, look for signs of **HIE** (lethargy, hypotonia, poor suck, seizures), birth trauma, anomalies, signs of sepsis.

APGAR score (at 1 and 5 minutes; every 5 min until 7 or above, up to 20 min)

Sign	0	1	2
Appearance (colour)	Blue or pale all over	Body pink, extremities blue	Pink all over
Pulse (heart rate)	Absent	Below 100	100 or above
Grimace (reflex irritability)	No response	Grimace	Cry, cough, sneeze
Activity (muscle tone)	Limp	Some flexion	Active movement
Respiration	Absent	Weak, irregular, gasping	Strong cry

- 7-10: normal; 4-6: moderately depressed; 0-3: severely depressed.
- **Never wait for the 1-minute Apgar to start resuscitation.** By 1 minute you should already be ventilating.
- The 5-minute score better reflects response to resuscitation; a low 5- or 10-minute score is associated with HIE and poor outcome.

After resuscitation: what each finding means

System	What to look for	Normal to document	Abnormal → action
Vital signs	RR, HR, temperature, SpO_2 (if oximeter)	RR 40–60, HR 100–160 (at rest), temperature 36.5–37.5°C, pre-ductal SpO_2 85–95% by 10 min, then ≥90% once stable	RR >60, grunting, indrawing → oxygen/CPAP; temperature <36.5 → skin-to-skin/warm; >37.5 → avoid overheating, think sepsis
Head	Caput, cephalhaematoma, boggy swelling crossing sutures	Caput only	**Subgaleal haemorrhage** (after vacuum): swelling crossing suture lines, falling Hb → urgent SCBU
Respiratory	Air entry, symmetry, meconium	Equal air entry	Asymmetry + sudden deterioration → pneumothorax; scaphoid abdomen → diaphragmatic hernia
Circulation	Capillary refill, pallor, pulses	CRT <3 s, pink	Pale, CRT ≥3 s after APH → hypovolaemia: saline 10 mL/kg
Neurology (Sarnat)	Alertness, tone, suck, Moro, seizures	Alert, flexed, good suck	Lethargy, hypotonia, weak suck = **moderate HIE**; stupor, flaccid = **severe** → SCBU
Birth trauma	Arm movement, clavicle	Symmetrical	Erb's palsy (after shoulder dystocia), clavicle fracture
Anomalies	Full top-to-toe examination	None	Refer as appropriate
Feeding	Rooting, attaching at the breast	Suckling well	Poor suck → expressed breast milk by cup/NGT, check glucose

Grading HIE (Sarnat, simplified): what you will see on the ward

Feature	Mild	Moderate	Severe
Level of consciousness	Hyper-alert, irritable	Lethargic	Stupor/coma

Feature	Mild	Moderate	Severe
Tone	Normal or increased	Hypotonia	Flaccid
Suck	Weak	Weak/absent	Absent
Seizures	None	Common	Common or suppressed
Outcome	Usually normal	Variable (risk of disability)	High mortality, disability

Bedside tests

Test	When	Normal / threshold	What you do
Heart rate (stethoscope; cord pulse)	After 1 min of ventilation, then every 1–2 min during resuscitation	**≥100** normal; **<100** slow; **<60** compressions (after 30 s of effective ventilation)	Drives each step of the algorithm
Pulse oximetry (right hand, pre-ductal)	Hospital, when ventilating or giving oxygen	60–65% at 1 min rising to **85–95% at 10 min**	Titrate oxygen; avoid hyperoxia
Temperature	On admission to the newborn corner and every 15–30 min after resuscitation	**36.5–37.5°C**	Hypothermia → skin-to-skin, hat, warm room; avoid overheating
Blood glucose (glucometer, heel prick)	All resuscitated babies, preterm/LBW, poor feeders, babies of diabetic mothers	Uganda/WHO practice: treat if **<2.6 mmol/L** (NICE uses 2.0 mmol/L in the first 48 h; check your unit protocol)	10% dextrose 2 mL/kg IV, then maintenance; or feed/EBM by NGT
Weight	After stabilisation	≥2,500 g	LBW → KMC, feeding plan, gentamicin dose adjusted
APGAR	1 and 5 min (every 5 min until ≥7, up to 20 min)	7–10	A low 5- or 10-min score predicts HIE; document
HIV DNA PCR (DBS)	HIV-exposed baby, per national schedule		Infant prophylaxis starts regardless of the result

UGANDA CONTEXT

Uganda and WHO practice treat neonatal glucose **below 2.6 mmol/L**; NICE (UK) uses **2.0 mmol/L** in the first 48 hours. Follow your unit protocol and say which threshold you are using.

Laboratory and imaging

Test	When	Expected / abnormal finding	How it changes management
Blood glucose (lab or glucometer)	Every resuscitated baby, repeated	<2.6 mmol/L (local threshold)	Dextrose bolus and infusion; frequent feeds. Glucometer at HC IV and hospitals (often absent at HC III: feed early and refer)
Blood culture	Sepsis risk factors or signs, before antibiotics	Growth	Guides antibiotic choice and duration. Regional/national referral hospitals only; do not delay antibiotics to get one
FBC (Hb, WCC, platelets)	Pale baby, suspected sepsis, subgaleal haemorrhage	Hb normally about 14–20 g/dL at birth; low Hb = blood loss; low platelets = sepsis/DIC	Transfusion; antibiotics. HC IV and hospitals

Test	When	Expected / abnormal finding	How it changes management
CRP	Suspected sepsis (serial at 24–48 h)	Rising CRP supports infection	Helps decide on stopping antibiotics after 48 h. Regional/national referral
Serum electrolytes, calcium, creatinine	Moderate/severe HIE, seizures, oliguria	Low sodium (SIADH), low calcium, raised creatinine (AKI)	Fluid restriction; correct calcium; adjust gentamicin. Regional/national referral
Bilirubin	Jaundice (more likely after bruising or sepsis)	Above phototherapy line for age	Phototherapy. SCBUs at Mengo, Mulago/Kawempe, regional referrals
Blood gas (cord or arterial)	Severe asphyxia at a tertiary centre	pH <7.0 or base deficit ≥12 = significant acidosis	Supports HIE diagnosis; rarely available in Uganda
Chest X-ray	Persistent respiratory distress, suspected pneumothorax, meconium aspiration, diaphragmatic hernia	Pneumothorax, patchy infiltrates, bowel in the chest	Chest drain, oxygen/CPAP, surgery. Hospitals with radiology
Cranial ultrasound	Preterm, suspected intracranial haemorrhage, severe HIE	IVH, oedema	Prognosis and counselling. National referral / some regional hospitals
Lumbar puncture	Seizures with suspected sepsis, once stable	Meningitis	Antibiotic dose and duration (meningitic doses)

7. Clinical Acumen

Red flags

RED FLAG

- **Not breathing or gasping** after drying and stimulation: ventilate now
- **Chest not rising** with ventilation: improve ventilation (mask, head, airway, mouth, pressure)
- **Heart rate below 100** after 1 minute of ventilation: call for help, continue ventilation, advanced care
- **Heart rate below 60** after 30 s of effective ventilation: chest compressions (where trained)
- After resuscitation: **seizures**, **poor suck**, **floppiness**, **apnoea**, **temperature below 36.5 or above 37.5**, **glucose below 2.6 mmol/L**, **respiratory distress** (grunting, severe indrawing, RR above 60): admit to SCBU/refer

RED FLAG

Gasping is not breathing. A baby who is gasping, floppy and pale at birth is in **secondary apnoea**: ventilate at once. Do not stop ventilating to listen to the heart yourself; the helper checks the heart rate.

Thinking like a clinician

- **"Gasping is not breathing."** A baby who gasps is in severe hypoxia; ventilate.
- **"No chest rise? It is almost always the mask seal or head position,"** not the bag. Fix technique before looking for other causes.
- **"Heart rate is the vital sign of the newborn."** A rising HR means ventilation is working even if the baby is not yet breathing.

- **"Do not waste time on suction."** Suction is for obstruction only; repeated suction delays ventilation and causes bradycardia.
- **"Oxygen is not a substitute for ventilation."** Blowing oxygen at a non-breathing baby does nothing; ventilation with room air saves lives.
- **"If the HR stays slow despite a good chest rise"**, think: pneumothorax, diaphragmatic hernia, hypovolaemia (pale, poor perfusion after APH), congenital heart disease, or too little time.
- **"The workup happens in three time zones"**: **before the birth** (anticipate and prepare), **at the birth** (the rapid assessment that drives the HBB Action Plan) and **after resuscitation** (a structured newborn assessment that decides whether the baby stays with the mother or goes to the SCBU). The first two are measured in seconds, so the order must be automatic.
- **Common student mistakes**: forgetting to dry and remove the wet towel; mask covering the eyes; over-extending the neck; ventilating too fast (over 60/min) or with big squeezes (lung injury); stopping ventilation to check the HR repeatedly; not calling for help early; forgetting the time of birth; not recording the resuscitation.

Putting it together

Model one-line summary: "A term male, 3.1 kg, born after prolonged second stage with thick meconium, **not breathing at birth**, ventilated from 50 s, HR 80 at 1 min of ventilation rising to 140 by 3 min, spontaneous breathing at 5 min, **APGAR 2, 6, 8**; now lethargic with a weak suck: **moderate perinatal asphyxia with possible moderate HIE**."

Problem list: (1) perinatal asphyxia / possible HIE; (2) possible meconium aspiration; (3) risk of hypoglycaemia, hypothermia and seizures; (4) sepsis risk (if ROM >18 h or maternal fever); (5) feeding difficulty; (6) communication with the mother.

Working diagnosis: name the **severity** (mild/moderate/severe HIE by Sarnat), the **complications** (seizures, aspiration, hypoglycaemia) and the **risk factors** (sepsis).

EXAM FAVOURITE

Worked example 1: scoring the APGAR. At 1 minute a baby is blue all over (0), HR 80 (1), grimaces to suction (1), some flexion (1), weak irregular breathing (1): **APGAR 4 at 1 min**. At 5 minutes: body pink with blue hands and feet (1), HR 140 (2), cries on stimulation (2), active movement (2), strong cry (2): **APGAR 9 at 5 min**. **Interpretation**: moderately depressed at 1 minute, good response by 5 minutes. **Key point**: the ventilation decision was made at 30–60 seconds on "not breathing", not on the APGAR.

EXAM FAVOURITE

Worked example 2: after resuscitation. A 2.2 kg baby born at 35 weeks needed 3 minutes of bag-and-mask ventilation. At 1 hour: temperature **35.8°C**, RR 64 with mild grunting, SpO_2 88% in air, glucose **2.1 mmol/L**, poor suck, mother had ROM for 30 hours. **Interpretation**: **hypothermia, hypoglycaemia, respiratory distress** and **sepsis risk** (prolonged ROM) in a late-preterm LBW baby after asphyxia. **Action**: skin-to-skin/KMC or warmer; **10% dextrose 2 mL/kg IV** then maintenance 60 mL/kg/day, recheck glucose in 30 min; oxygen (CPAP if available) to SpO_2 90–95%; blood culture if available, then **ampicillin 50 mg/kg 12-hourly + gentamicin** (3 mg/kg 24-hourly for LBW per local protocol); SCBU admission (from an HC III/IV: refer with the mother, baby kept warm).

EXAM FAVOURITE

Worked example 3: heart rate that won't rise. In a hospital, a term baby is ventilated with good chest rise for 30 s but the HR is **50**. **Interpretation**: HR <60 despite effective ventilation → **chest compressions 3:1** with 100% oxygen, recheck after 60 s; if still <60, **adrenaline 1:10,000 0.1–0.3 mL/kg IV** via UVC. If the chest had **not** been rising, the right answer would be **improve ventilation first** (MR SOPA), not compressions.

8. Complications and Troubleshooting

Troubleshooting during resuscitation

Problem	Likely cause	Fix
Chest not rising	Mask leak, head position, secretions, closed mouth, too little pressure	**MR SOPA**: reapply mask, reposition, suction, open mouth, more pressure, alternative airway
HR stays slow despite good chest rise	Pneumothorax, diaphragmatic hernia, hypovolaemia, congenital heart disease, too little time	Continue ventilation; compressions if <60 after 30 s; volume if pale after APH; chest X-ray/transillumination in hospital
Sudden deterioration, asymmetric chest	Pneumothorax from excessive pressure	Senior; needle decompression/chest drain in hospital
Bag does not re-inflate or no pressure on testing	Faulty or wrongly reassembled bag	Replace; test every bag before every birth
Baby breathing but cold	Wet towel left on, no hat, draughts	Dry, remove wet cloth, hat, skin-to-skin, warm room
Bradycardia during suction	Vagal stimulation, deep suction	Stop suction; ventilate

Post-resuscitation care

The baby who needed ventilation is at risk of **hypoxic-ischaemic injury, hypoglycaemia, hypothermia, seizures, sepsis, feeding problems and renal/liver dysfunction**.

1. **Short ventilation, now breathing well, HR normal, good tone**: keep **skin-to-skin with the mother**; monitor breathing, HR, colour and temperature **every 15 minutes for the first 2 hours**, then regularly for 24 hours; breastfeed early; do not discharge early.
2. **Prolonged ventilation, abnormal tone, poor suck, seizures, or ongoing respiratory distress**: **admit to SCBU** (or refer after stabilisation from HC III/IV) keeping the baby warm (skin-to-skin/KMC during transfer), with a referral note.
3. **Temperature**: keep 36.5-37.5°C; **avoid hyperthermia** (worsens brain injury). (Where available, therapeutic hypothermia for moderate/severe HIE only within a proper protocol; not standard in most Ugandan units.)
4. **Glucose**: check blood glucose; treat if below **2.6 mmol/L** (see table).
5. **Fluids and feeding**: breastfeed if able; if not able to suck, expressed breast milk by cup or nasogastric tube; if unable to feed enterally, **IV 10% dextrose 60 mL/kg/day on day 1** (increase daily), restricting fluids if poor urine output.
6. **Seizures**: **phenobarbital** (see table); check glucose, calcium where possible, and treat sepsis.
7. **Infection**: if risk factors (prolonged ROM, maternal fever, foul liquor) or signs of sepsis: **ampicillin + gentamicin** after blood culture where available.
8. **Monitor urine output**, feeding, neurological state; plan follow-up for development.

9. **Communicate** honestly with the mother and family; **document** (section 9).

Drug table

Drug	Dose	Route	Frequency	Notes
Adrenaline 1:10,000	0.01-0.03 mg/kg (0.1-0.3 mL/kg)	IV/IO (UVC)	Every 3-5 min if HR below 60	ET: 0.05-0.1 mg/kg (0.5-1 mL/kg) while getting IV access
Normal saline	10 mL/kg	IV over 5-10 min	May repeat once	Only if hypovolaemia suspected
10% dextrose (hypoglycaemia)	2 mL/kg bolus	IV	Once, then maintenance infusion	Recheck glucose in 30 min; target 2.6 mmol/L or above. If no IV access: breast milk/10% dextrose by NGT
10% dextrose (maintenance)	60 mL/kg/day (day 1)	IV	Continuous	Increase by about 20 mL/kg/day as tolerated
Phenobarbital (seizures)	Loading **20 mg/kg**	IV slowly over 15-20 min (or IM)	Once; extra 10 mg/kg if seizures persist (to 40 mg/kg total)	Maintenance **5 mg/kg/day** once daily
Ampicillin	50 mg/kg	IV/IM	Every 12 h (first week of life); every 8 h thereafter	With gentamicin, for suspected sepsis
Gentamicin	5 mg/kg (term)	IV/IM	Every 24 h	3 mg/kg every 24 h in low birth weight/preterm per local protocol
Vitamin K	1 mg (term), 0.5 mg (preterm)	IM	Once	As for all newborns

UGANDA CONTEXT

At an HC III/HC IV without an SCBU, after resuscitation: keep the baby skin-to-skin, give the first dose of antibiotics if indicated, check glucose (or give an early feed/expressed breast milk), and **refer early** to the hospital with the mother, keeping the baby warm during transport. Do not transfer a baby who is still being bag-ventilated without a health worker continuing ventilation throughout.

Complications

Complication	Notes
HIE, seizures, cerebral palsy, developmental delay	The goal of rapid ventilation is to prevent this
Pneumothorax	From excessive pressure/volume; sudden deterioration, asymmetrical chest
Hypothermia, **hypoglycaemia**	Common after resuscitation; check and prevent
Aspiration (meconium aspiration syndrome)	Respiratory distress, may need oxygen/CPAP
Multi-organ dysfunction	Oliguria (acute kidney injury), liver dysfunction, NEC, coagulopathy
Infection	From risk factors or equipment (dirty bags, masks)
Trauma from rough stimulation or deep suction	Bradycardia, laryngospasm, mucosal injury
Maternal psychological trauma	Communicate, support, and debrief the family and staff

9. Documentation and Presenting Findings

What to record

After resuscitation, record exactly: time of birth, time ventilation started, how long, heart rate response, when spontaneous breathing started, APGARs, drugs, and who was present. This record is used in SCBU and in perinatal death reviews. Add the post-resuscitation examination findings, temperature, glucose, the plan (stay with mother or SCBU/referral) and what the family was told.

Model presentation

"Baby of Mrs R.K., a male infant born at 38 weeks by spontaneous vaginal delivery at 02:15 after prolonged second stage with **thick meconium-stained liquor** and fetal bradycardia. He was **not breathing and floppy after drying and stimulation**; mouth and nose were cleared of meconium, and **bag-and-mask ventilation in room air was started at 50 seconds**. After 1 minute of ventilation with good chest rise, **heart rate was 80**; ventilation continued, and by 3 minutes the heart rate was 140 and he began to breathe spontaneously at 5 minutes. **Apgar scores 2, 6 and 8** at 1, 5 and 10 minutes. Birth weight 3.1 kg. Now in the SCBU, day 1: temperature 36.8, RR 56, mild subcostal recession, **lethargic with a weak suck**, no seizures, glucose 3.4 mmol/L. **Assessment: moderate perinatal asphyxia (possible moderate HIE) with possible meconium aspiration.** Plan: oxygen to keep SpO_2 91-95%, **IV 10% dextrose 60 mL/kg/day**, NGT feeds of expressed breast milk as tolerated, **ampicillin and gentamicin** after blood culture, watch for seizures (phenobarbital 20 mg/kg if they occur), monitor glucose, temperature, urine output, and counsel the mother."

10. OSCE Checklist

Station: HBB on the NeoNatalie simulator

Step	Done well when	Critical?
Prepare	Identifies helper, prepares warm area, checks and **tests** bag and mask, lays out equipment	Yes
Time of birth	Notes and says the time of birth	Yes
Dry	**Dries thoroughly**, removes wet cloth, covers and keeps warm	Yes
Assess	Assesses crying/breathing	Yes
Airway	Positions head (slightly extended); **clears airway only if needed**	No
Stimulate	Rubs the back 2–3 times; no slapping	No
Ventilate by 60 s	Recognises not breathing; **cuts cord and starts ventilation within 60 s**	Yes
Mask	Correct size and seal (chin, mouth, nose; not eyes); C-grip with jaw lift	Yes
Rate and chest rise	Ventilates at **40/min** with visible chest rise	Yes
Improve ventilation	Reapplies mask, repositions head, clears airway, opens mouth, increases pressure when the chest does not rise	Yes
Heart rate	Calls for help; asks helper to **check heart rate** after 1 min of ventilation without stopping ventilation	Yes
Interpret	Correctly interprets **normal vs slow HR** and continues/decides on advanced care	Yes
Stop and monitor	Stops ventilation when the baby breathes well; monitors with mother	No
Communicate and document	Communicates with the mother; documents	No

Step	Done well when	Critical?
Decontaminate	Decontaminates and tests equipment after use	No

11. Key Points

High-yield points

- **Ventilation is the key**: about 10% need help, 3–6% need bag-and-mask, <1% need compressions or drugs.
- **Golden Minute**: ventilating within 60 s of birth.
- **Prepare before every birth**: helper, warm area, tested bag, masks size 1 and 0.
- **Gasping is not breathing.**
- Effective ventilation = **gentle chest rise at 40/min**; room air for term babies.
- No chest rise: **MR SOPA** before anything else.
- **Heart rate** tells you if ventilation is working: ≥100 normal; <60 after 30 s of effective ventilation → compressions 3:1.
- Suction only if the airway is blocked; no routine tracheal suction for meconium.
- Neonatal adrenaline is **1:10,000**.
- After resuscitation: temperature, glucose, breathing and tone every 15 min for 2 h; SCBU if abnormal.
- APGAR is recorded, never used to decide to start.

Numbers & doses

Item	Value
Babies needing help to breathe / bag-mask ventilation	About **10%** / about **3–6%**; compressions or drugs **<1%**
Golden Minute	Ventilation started **within 60 s** of birth
Ventilation rate	**40 breaths/min** (30–60 acceptable), "squeeze-two-three"
Mask size	**Size 1 term**, **size 0 preterm/small**
Starting gas	**Room air (21%)** for term and ≥35 weeks; 21–30% below 35 weeks
HR thresholds	**≥100 normal**; **<100** continue and improve ventilation, call help; **<60** after 30 s of effective ventilation → compressions
Compressions	Two-thumb, lower third of sternum, depth **one-third of AP chest**, **3:1** (90 compressions + 30 breaths/min), 100% O_2, recheck after **60 s**
Adrenaline 1:10,000 (0.1 mg/mL)	**0.01–0.03 mg/kg IV (0.1–0.3 mL/kg)**, every 3–5 min; ET **0.05–0.1 mg/kg (0.5–1 mL/kg)**
Making 1:10,000	**1 mL of 1:1,000 + 9 mL normal saline**
Volume	**Normal saline 10 mL/kg** over 5–10 min
Pre-ductal SpO_2 targets	60–65% at 1 min → **85–95% at 10 min**
Normal temperature	**36.5–37.5°C**
Hypoglycaemia	Treat if **<2.6 mmol/L** (local; NICE 2.0) → **10% dextrose 2 mL/kg IV**
Day-1 maintenance fluid	**10% dextrose 60 mL/kg/day**
Phenobarbital (seizures)	**20 mg/kg** IV load (+10 mg/kg to 40 mg/kg total); maintenance **5 mg/kg/day**
Ampicillin / gentamicin	**50 mg/kg** 12-hourly (first week) / **5 mg/kg** 24-hourly (3 mg/kg LBW)

Item	Value
Vitamin K	**1 mg IM** term; 0.5 mg preterm
Delayed cord clamping	**1–3 min** in breathing babies
Post-resuscitation observations	**Every 15 min for 2 h**, then regularly for 24 h
Stop resuscitation (WHO)	**No HR after 10 min** of effective ventilation; **HR <60 and no breathing after 20 min**
APGAR timing	**1 and 5 min** (every 5 min until ≥7, up to 20 min)

Exam traps

- **"Wait for the 1-minute APGAR before starting resuscitation."** → Wrong. By 1 minute a non-breathing baby should already be **ventilated**.
- **"Suction every baby, especially through meconium."** → Wrong. Suction **only if the airway is blocked**; no routine tracheal suction for meconium in a non-vigorous baby: **ventilate**.
- **"The baby is gasping, so he is breathing."** → Wrong. **Gasping = not breathing**: ventilate.
- **"HR is 50: start chest compressions immediately."** → Only after **30 s of ventilation that moves the chest**. If the chest is not rising, **fix ventilation first** (MR SOPA).
- **"Give oxygen by face mask to a non-breathing baby."** → Wrong. Blow-by oxygen does nothing; **ventilate with room air**.
- **"Use 1:1,000 adrenaline at 0.1 mL/kg."** → Dangerous tenfold overdose. Neonatal adrenaline is **1:10,000**.
- **"Rub, slap, hold upside down."** → Wrong. Stimulate by **rubbing the back 2–3 times** or flicking the soles.
- **"Baby breathing now, so he can go home tomorrow like any other."** → Wrong. A ventilated baby needs **temperature, glucose, breathing and neurological monitoring** (every 15 min for 2 h, then for 24 h) and SCBU if not normal.

Questions seniors ask

Viva questions

1. **What is the Golden Minute?** The first 60 seconds after birth, by the end of which a non-breathing baby should be receiving bag-and-mask ventilation.
2. **What rate do you ventilate at?** 40 breaths per minute (30-60 acceptable), squeeze-two-three.
3. **How do you know ventilation is effective?** Gentle chest rise with each breath; rising heart rate.
4. **When do you start chest compressions?** HR below 60 after at least 30 s of effective ventilation; ratio 3:1; 100% oxygen.
5. **Adrenaline dose?** 0.01-0.03 mg/kg IV of 1:10,000 (0.1-0.3 mL/kg).
6. **When do you suction?** Only if secretions or meconium obstruct the airway (and the baby is not crying); mouth before nose.
7. **Why is Apgar not used to start resuscitation?** It is only assigned at 1 minute, by which time ventilation should already have started.
8. **What do you do for a vigorous baby born through meconium?** Routine care; no suction needed.

On the ward round

1. **"What did you do in the first 30 seconds?"** Noted the time, dried thoroughly, removed the wet towel, kept warm, assessed crying and breathing.
2. **"When did you start ventilating, and how do you know it worked?"** Before 60 s; gentle chest rise with each breath and a **rising heart rate**.
3. **"The chest isn't rising. What next?"** Reapply the mask, reposition the head, clear the airway, open the mouth, squeeze harder (MR SOPA; alternative airway in hospital).

4. **"What is his heart rate and how did you measure it?"** Stethoscope over the left chest, count for 6 s × 10 (the cord pulse can underestimate).
5. **"What are you worried about in the first 24 hours after resuscitation?"** HIE and seizures, hypoglycaemia, hypothermia, sepsis, feeding difficulty, respiratory distress (meconium aspiration, pneumothorax).
6. **"How would you grade his encephalopathy?"** Sarnat: mild (hyper-alert, normal tone), moderate (lethargic, hypotonic, weak suck, seizures), severe (stupor, flaccid, absent suck).
7. **"When would you stop?"** No heart rate after 10 minutes of effective ventilation, or HR <60 without breathing after 20 minutes (WHO), with a senior, and with honest, supportive communication with the family.

Mnemonics

- **HBB flow**: "**Dry, Cry? Clear, Stimulate, Breathe? Ventilate by 60!**"
- **Improve ventilation (NRP)**: "**MR SOPA**" = Mask adjust, Reposition airway, Suction, Open mouth, Pressure increase, Alternative airway.
- **Ventilation rhythm**: "**Squeeze**-two-three" = 40/min.
- **Compressions**: "**One-and-two-and-three-and-breathe**" = 3:1, 120 events/min (90 compressions + 30 breaths).
- **Heart rate thresholds**: "**100 is normal, 60 is compressions**".
- **APGAR**: Appearance, Pulse, Grimace, Activity, Respiration.
- **Adrenaline**: "**0.1 to 0.3 mL/kg IV of 1 in 10,000**" ("point-one to point-three").

OSCE tips

- Talk out loud: "I am drying… the baby is not crying… I am clearing the airway because I see secretions… stimulating… still not breathing… I am cutting the cord and starting ventilation at 45 seconds."
- Look at the **clock** at the start and say the time. Examiners check that ventilation starts **before 60 seconds**.
- Keep ventilating while the helper checks the heart rate. Do not stop to listen yourself.
- Test the bag against your palm at the start and decontaminate it at the end; both are on the checklist.

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SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	What is the Golden Minute?	The first **60 seconds after birth**: a baby not breathing after drying and stimulation must be receiving **bag-and-mask ventilation** by then.
2	Roughly what proportion of newborns need help to breathe, and what proportion need bag-and-mask ventilation?	About **10%** need some help; about **3-6%** need bag-and-mask ventilation; under 1% need compressions or drugs.
3	First two actions for every baby at birth (HBB)?	**Note the time of birth** and **dry thoroughly** (then remove the wet cloth and keep warm).
4	When should you suction a newborn's airway?	**Only if the airway is blocked** by secretions, or meconium-stained liquor and the baby is **not crying**. Mouth first, then nose, not deep.
5	How do you stimulate a baby to breathe?	**Rub the back** firmly 2-3 times (or flick the soles). Never slap, shake, hang upside down or squeeze the chest.
6	Bag-and-mask ventilation rate in HBB?	**40 breaths per minute** (30-60 acceptable): 'Squeeze-two-three'.
7	Mask sizes for term and preterm babies?	**Size 1** for term; **size 0** for small/preterm. It covers chin, mouth and nose but **not the eyes**.
8	How do you know ventilation is effective?	**Gentle chest rise** with each breath, then a **rising heart rate**.
9	Five steps to improve ventilation (HBB)?	1. **Reapply mask** 2. **Reposition head** (slight extension) 3. **Clear airway** 4. **Open mouth** slightly 5. **Squeeze bag harder**.
10	Heart rate thresholds in neonatal resuscitation?	**100 or above = normal**; **below 100 = slow** (continue/improve ventilation, call for help); **below 60 after 30 s of effective ventilation = chest compressions** (advanced).
11	How do you check a newborn's heart rate quickly?	Stethoscope over the left chest (or feel the base of the cord): count for **6 seconds and multiply by 10**. Helper does it while you keep ventilating.
12	Chest compression technique in newborns?	**Two-thumb** technique on the **lower third of the sternum**, depth **one-third of AP chest diameter**, **3:1** ratio (90 compressions + 30 breaths/min), 100% oxygen.
13	Neonatal adrenaline dose and strength?	**1:10,000 (0.1 mg/mL)**: **0.01-0.03 mg/kg IV (0.1-0.3 mL/kg)** via UVC, every 3-5 min; ET 0.05-0.1 mg/kg (0.5-1 mL/kg).
14	Oxygen to start ventilation in a term baby?	**Room air (21%)**; titrate to pre-ductal SpO2 targets. Do not delay ventilation to find oxygen. Increase to 100% if chest compressions are needed.
15	APGAR components and scoring?	**A**ppearance, **P**ulse, **G**rimace, **A**ctivity, **R**espiration; each 0-2; at 1 and 5 min (then every 5 min until 7 or above). **Not used to decide to start resuscitation.**
16	Management of a non-vigorous baby born through meconium?	Clear mouth/nose if blocked, then **start ventilation within 1 minute**. **No routine tracheal suction/intubation.**
17	When may resuscitation be stopped (WHO)?	**No heart rate after 10 minutes** of effective ventilation; or **HR below 60 with no spontaneous breathing after 20 minutes**. Senior decision.

#	Question	Answer
18	Primary vs secondary apnoea?	**Primary**: responds to stimulation. **Secondary (terminal)**: follows gasping, does **not** respond to stimulation, needs ventilation. At birth you cannot tell them apart, so ventilate if not breathing after stimulation.
19	How do you test a self-inflating bag before a birth?	Seal the mask (or the outlet) against your palm and squeeze: you should **feel pressure** and the **pressure-release valve** should open; the bag should **re-inflate** quickly.
20	Neonatal hypoglycaemia threshold and emergency treatment?	Glucose **below 2.6 mmol/L**: **10% dextrose 2 mL/kg IV**, then maintenance infusion; if no IV, breast milk/dextrose by NGT. Recheck in 30 min.
21	First-line drug and dose for neonatal seizures after asphyxia?	**Phenobarbital 20 mg/kg IV** slowly (or IM) loading; maintenance **5 mg/kg/day**. Check glucose too.
22	Is gasping 'breathing' in HBB?	**No.** Gasping is a sign of severe hypoxia: treat as not breathing and **ventilate**.
23	Score this 1-minute APGAR: blue all over, HR 80, grimace, some flexion, weak irregular breathing.	0 + 1 + 1 + 1 + 1 = **4** (moderately depressed). But ventilation should already have started on "not breathing" before this score was assigned.
24	Which checks do you do on every baby who needed resuscitation, and how often?	**Breathing, HR, colour, temperature (36.5–37.5°C), tone/suck, and blood glucose**; every **15 min for the first 2 h**, then regularly for 24 h. SCBU if lethargic, poor suck, seizures, distress or abnormal glucose/temperature.
25	HR is 50 but the chest is NOT rising with ventilation. Next step?	**Improve ventilation first** (MR SOPA: mask, reposition, suction, open mouth, pressure, alternative airway). Compressions only after **30 s of ventilation that moves the chest** with HR still <60.
26	Late-preterm baby after resuscitation: temperature 35.8°C, glucose 2.1 mmol/L, ROM 30 h. Three immediate actions?	**Warm** (skin-to-skin/KMC or warmer); **10% dextrose 2 mL/kg IV** then 60 mL/kg/day and recheck in 30 min; **ampicillin + gentamicin** (after blood culture if available) for sepsis risk; admit/refer to SCBU.
27	Which investigations help after birth asphyxia, and which are usually available only at referral hospitals?	Everywhere with a glucometer: **glucose**; HC IV/hospitals: **FBC**; regional/national referral: **blood culture, CRP, electrolytes/calcium/creatinine, bilirubin, chest X-ray, cranial ultrasound**; blood gas rarely available.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A term baby is born at an HC III after a normal labour. After thorough drying he is not crying, is floppy and is not breathing. There are no secretions visible. The time is 30 seconds after birth. What is the NEXT step?

A Rub the back to stimulate breathing, and if still not breathing, clamp and cut the cord and start bag-and-mask ventilation by 60 seconds
B Deep suction of the oropharynx and nose for 30 seconds
C Hold the baby upside down and slap the buttocks
D Wait for the 1-minute Apgar score before deciding
E Give oxygen by nasal prongs and observe

2. A student starts bag-and-mask ventilation on an apnoeic newborn but the chest is not rising. The mask is size 1, and the head is flexed onto the chest. What is the FIRST corrective step?

A Increase the rate to 80 breaths per minute
B Start chest compressions
C Give adrenaline
D Reapply the mask for a better seal and reposition the head into slight extension
E Stop ventilating and suction for 1 minute

3. After 1 minute of ventilation with good chest rise, a helper counts the heart rate at 8 beats in 6 seconds. The baby is still not breathing. What is the MOST appropriate action?

A Stop ventilation and start chest compressions immediately
B Continue ventilation, call for help, keep improving ventilation and decide on advanced care
C Stop resuscitation because the baby is not responding
D Give naloxone
E Stop ventilation and stimulate again

4. In Mengo's labour ward, a baby born through thick meconium-stained liquor is limp and not breathing at birth. The team includes a nurse skilled in intubation. What is the BEST initial approach according to current guidance?

A Routinely intubate and suction the trachea before any breaths are given
B Suction the mouth and nose deeply for at least 1 minute
C Avoid ventilation until the meconium is completely removed
D Give adrenaline first
E Dry, keep warm, clear the mouth and nose if obstructed, and start bag-and-mask ventilation within the first minute

5. A term baby in the SCBU has had 30 seconds of ventilation via a well-sealed mask with good chest rise, but the heart rate is 50 bpm. What is the NEXT step? Advanced

A Continue ventilation alone for 5 more minutes
B Give 10 mL/kg normal saline
C Start chest compressions at a 3:1 ratio with ventilation, increase oxygen to 100%
D Stop resuscitation
E Give 10% dextrose 2 mL/kg

6. A 3 kg baby remains at a heart rate of 40 bpm despite 60 seconds of well-coordinated compressions and ventilation with 100% oxygen. An umbilical venous catheter is in place. Only adrenaline 1:1,000 ampoules are available. What is the CORRECT dose? Advanced

- **A** Dilute 1 mL of 1:1,000 with 9 mL of normal saline to make 1:10,000 and give 0.3-0.9 mL IV
- **B** 3 mL of 1:1,000 IV
- **C** 0.1 mL of undiluted 1:1,000 IM
- **D** 1 mg IV bolus
- **E** 3 mL of 1:10,000 endotracheally

7. A baby at an HC IV is ventilated for 3 minutes and then breathes spontaneously with a heart rate of 150. Tone improves. What is the BEST plan for the next few hours?

- **A** Discharge home with the mother after 2 hours
- **B** Keep skin-to-skin with the mother, monitor breathing, heart rate, colour and temperature every 15 min for 2 h, start breastfeeding, check glucose if poor feeding, and observe for at least 24 h
- **C** Separate from the mother and place in a cot under a window
- **D** Bathe the baby immediately to remove meconium and blood
- **E** Give prophylactic phenobarbital

8. A baby is born with a heart rate of 90, weak irregular breathing, some flexion of the limbs, a grimace on suction and a pink body with blue extremities. What is the Apgar score?

- **A** 3
- **B** 7
- **C** 6
- **D** 5
- **E** 4

9. Which piece of equipment is MOST essential to have tested and ready at every birth in a Ugandan HC III?

- **A** A pulse oximeter
- **B** An oxygen cylinder with a blender
- **C** A laryngoscope and endotracheal tubes
- **D** An umbilical venous catheter
- **E** A self-inflating neonatal bag with size 0 and 1 masks

10. A baby has been resuscitated with effective ventilation, compressions and adrenaline at a referral hospital. After 10 minutes there is still no detectable heart rate. What is the MOST appropriate course?

- **A** Continue indefinitely until the family arrives
- **B** Give sodium bicarbonate and continue for 1 hour
- **C** With the senior clinician, consider stopping resuscitation, then support and counsel the parents and document
- **D** Stop at 2 minutes because the baby was already dead
- **E** Transfer to another hospital with ongoing compressions

11. A preterm baby at 32 weeks (1.5 kg) is born at Mengo and is breathing irregularly with poor tone after drying. Which approach is MOST appropriate? Advanced

A Size 1 mask with 100% oxygen and high pressures

B Keep warm (plastic wrap/hat, radiant warmer), size 0 mask, start ventilation with 21-30% oxygen, and titrate to SpO2 targets; consider CPAP

C Bathe to remove vernix and then assess

D Wait 2 minutes to see if breathing improves

E Suction deeply to stimulate breathing

12. A term baby is ventilated with good chest rise and the heart rate rises to 140, but after 5 minutes the baby suddenly deteriorates, becomes cyanosed and bradycardic, and breath sounds are reduced on the right side. What is the MOST likely cause? Advanced

A Hypoglycaemia

B Congenital heart disease

C Normal transition

D Pneumothorax

E Hypothermia

13. A 6-hour-old baby who needed 4 minutes of bag-and-mask ventilation is lethargic, feeding poorly, and has a blood glucose of 1.9 mmol/L. There is IV access. What is the MOST appropriate management?

A Give 10% dextrose 2 mL/kg IV, then start maintenance 10% dextrose, and recheck glucose in 30 minutes

B Give 50% dextrose 5 mL/kg IV

C Give oral water and recheck tomorrow

D Give phenobarbital 20 mg/kg

E No treatment; glucose below 2.0 mmol/L is normal at 6 hours

14. The mother received pethidine 100 mg IM one hour before delivery. The baby is born apnoeic after drying and stimulation. What is the MOST appropriate first management?

A Naloxone IM before any other intervention

B Oxygen by face mask without ventilation

C Wait for the effect of pethidine to wear off

D Check blood glucose first

E Bag-and-mask ventilation within the Golden Minute, following the standard algorithm

15. A crying, vigorous term baby is born through thin meconium-stained liquor. The student reaches for the suction bulb. What should the supervisor advise?

A Suction the trachea with a catheter

B Suction the mouth and nose deeply before the first cry

C No suction is needed: dry, keep warm skin-to-skin, delay cord clamping and give routine care

D Take the baby immediately to the resuscitation table for observation

E Clamp the cord immediately and give oxygen

Answers & explanations

1. A · Rub the back to stimulate breathing, and if still not breathing, clamp and cut the cord and start bag-and-mask ventilation by 60 seconds

Per HBB: keep warm, clear airway **only if needed** (no secretions here), **stimulate**, and if not breathing **ventilate within the Golden Minute**. Deep suction (B) is not indicated without obstruction and causes bradycardia. Slapping or holding upside down (C) is harmful. Waiting for the Apgar (D) wastes the Golden Minute. Oxygen without ventilation (E) does not help an apnoeic baby.

2. D · Reapply the mask for a better seal and reposition the head into slight extension

No chest rise is almost always due to a **poor mask seal or head position**. The HBB steps are: reapply mask, reposition head, clear airway, open mouth, squeeze harder. Increasing the rate (A) does not fix an ineffective breath. Compressions (B) and adrenaline (C) are useless without effective ventilation. Prolonged suction (E) delays ventilation.

3. B · Continue ventilation, call for help, keep improving ventilation and decide on advanced care

8 in 6 s = **80 bpm (slow, below 100)**. HBB: **continue ventilation**, call for help, improve ventilation, advanced care. Compressions (A) are for HR **below 60** after 30 s of effective ventilation. Stopping (C) is far too early (the 10-minute rule applies to no heart rate). Naloxone (D) is not routinely recommended. Stopping to stimulate (E) interrupts the key intervention.

4. E · Dry, keep warm, clear the mouth and nose if obstructed, and start bag-and-mask ventilation within the first minute

Current ILCOR/AHA 2020 and HBB guidance: **no routine tracheal suctioning** for non-vigorous meconium babies; clear the airway if obstructed and **start ventilation within 1 minute**. Routine intubation and suction (A) delays ventilation without improving outcomes (it is reserved for suspected airway obstruction). Prolonged suction (B) and withholding ventilation (C) cause worsening hypoxia. Adrenaline (D) is never the first step.

5. C · Start chest compressions at a 3:1 ratio with ventilation, increase oxygen to 100%

HR below 60 after at least 30 s of effective ventilation = start **chest compressions** (two-thumb, lower third of sternum, 3:1) and increase oxygen to 100%; consider an advanced airway. Ventilation alone (A) is insufficient at this HR. Volume (B) is for suspected hypovolaemia. Stopping (D) is premature. Dextrose (E) does not treat bradycardia.

6. A · Dilute 1 mL of 1:1,000 with 9 mL of normal saline to make 1:10,000 and give 0.3-0.9 mL IV

Neonatal IV adrenaline is **0.01-0.03 mg/kg** of **1:10,000** (0.1-0.3 mL/kg), i.e. **0.3-0.9 mL** for 3 kg. Dilute 1:1,000 first. 3 mL of 1:1,000 (B) = 3 mg, a massive overdose. IM (C) is not used in resuscitation. 1 mg (D) is an adult dose. 3 mL of 1:10,000 ET (E) is 0.3 mg (0.1 mg/kg), at the top of the endotracheal range, but the ET route is less reliable and is only a stopgap when there is no IV access; with a UVC in place, give it IV.

7. B · Keep skin-to-skin with the mother, monitor breathing, heart rate, colour and temperature every 15 min for 2 h, start breastfeeding, check glucose if poor feeding, and observe for at least 24 h

A baby who needed ventilation is at risk of **hypothermia, hypoglycaemia, apnoea and HIE**: keep warm skin-to-skin, monitor closely, feed early, and observe for at least 24 h. Early discharge (A) risks missed deterioration. Separation into a cold cot (C) causes hypothermia. Early bathing (D) causes heat loss. Prophylactic phenobarbital (E) is not recommended; treat seizures if they occur.

8. D · 5

Appearance (pink body, blue extremities) **1**, Pulse (below 100) **1**, Grimace **1**, Activity (some flexion) **1**, Respiration (weak, irregular) **1** = **5**. Scores of 3 (A), 4 (E), 6 (C) and 7 (B) miscount at least one component. Remember that a baby with irregular breathing and HR below 100 should already be receiving help; the Apgar is recorded, not used to decide.

9. E · A self-inflating neonatal bag with size 0 and 1 masks

Ventilation with a **bag and mask** (room air) is the life-saving intervention in basic resuscitation, and HBB requires it at every birth. Pulse oximetry (A), blended oxygen (B), intubation equipment (C) and UVC (D) are advanced hospital tools; they are helpful but not essential for basic resuscitation, and their absence must never delay ventilation.

10. C · With the senior clinician, consider stopping resuscitation, then support and counsel the parents and document

WHO: in newborns with **no detectable heart rate after 10 minutes of effective ventilation**, resuscitation should be stopped; this is a **senior** decision, followed by compassionate communication and documentation. Indefinite resuscitation (A) or bicarbonate for an hour (B) is futile and not recommended. Stopping at 2 minutes (D) is premature. Transfer with ongoing compressions (E) is futile and unsafe.

11. B · Keep warm (plastic wrap/hat, radiant warmer), size 0 mask, start ventilation with 21-30% oxygen, and titrate to SpO2 targets; consider CPAP

Preterm babies lose heat fast and need **warmth**, a **size 0 mask**, **lower oxygen (21-30%)** titrated to SpO2, and gentle pressures; CPAP helps with surfactant deficiency. Size 1 and high oxygen/pressures (A) risk poor seal, oxygen toxicity and lung injury. Bathing (C) causes hypothermia. Waiting (D) wastes the Golden Minute. Deep suction (E) causes bradycardia and trauma.

12. D · Pneumothorax

Sudden deterioration during positive-pressure ventilation with **unilateral reduced breath sounds** suggests a **pneumothorax** (barotrauma). Needs needle aspiration by a skilled clinician. Hypoglycaemia (A) and hypothermia (E) do not give focal chest signs. Congenital heart disease (B) would not cause a sudden unilateral change. This is not normal transition (C).

13. A · Give 10% dextrose 2 mL/kg IV, then start maintenance 10% dextrose, and recheck glucose in 30 minutes

Glucose below **2.6 mmol/L** in a symptomatic baby: **10% dextrose 2 mL/kg IV**, then maintenance (60 mL/kg/day on day 1) and recheck. 50% dextrose (B) is hyperosmolar and damages veins/brain. Water (C) does not treat hypoglycaemia and delays care. Phenobarbital (D) is for seizures, not hypoglycaemia (though you must check glucose in any seizing baby). A glucose of 1.9 in a symptomatic baby is not acceptable (E).

14. E · Bag-and-mask ventilation within the Golden Minute, following the standard algorithm

Opioid-induced respiratory depression is treated with **ventilation**; the standard algorithm applies. Naloxone (A) is **not recommended** as part of initial resuscitation and must never delay ventilation (it can also precipitate withdrawal). Free-flow oxygen (B) does not help an apnoeic baby. Waiting (C) causes hypoxic injury. Glucose (D) is checked after stabilisation.

15. C · No suction is needed: dry, keep warm skin-to-skin, delay cord clamping and give routine care

A **vigorous** baby does not need suction, even with meconium; routine care (drying, skin-to-skin, delayed clamping, breastfeeding) is best. Tracheal (A) or deep (B) suction can cause bradycardia, laryngospasm and trauma. Separating the baby (D) and immediate clamping with oxygen (E) are unnecessary and interrupt normal transition.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Not breathing at an HC IV after prolonged second stage

At an HC IV, Mrs P.N., 20, primigravida at 40 weeks, has had a 3-hour second stage with FHR dropping to 100 in the last 20 minutes. The midwife calls you as the head crowns.

1. What do you do before the baby is born?

> Identify yourself as the **helper**; prepare a warm area and resuscitation surface; lay out 2-3 dry towels, hat, **bag and masks size 0 and 1**, suction bulb, stethoscope, clock, cord clamps and scissors; **test the bag and mask**. Anticipate asphyxia (prolonged second stage, fetal bradycardia).

2. The baby is born, limp, not crying. What are your first 30 seconds?

> **Note the time. Dry thoroughly**, remove the wet cloth, cover, hat, keep warm. Not crying: position the head slightly extended, **clear the airway only if secretions are blocking it**, **stimulate** by rubbing the back.

3. At 40 seconds the baby is still not breathing. What now?

> **Clamp and cut the cord**, move to the ventilation surface, apply a **size 1 mask** (chin, mouth, nose), C-grip with jaw lift, and **ventilate at 40/min** in room air, watching for **chest rise**; improve ventilation if needed (mask, head, airway, mouth, squeeze harder).

4. After 1 minute of ventilation with good chest rise, the baby is not breathing; HR 70. What next?

> **Call for help**, **continue ventilation**, keep improving ventilation, reassess HR. At an HC IV without advanced skills, keep ventilating (HR is not below 60); if a trained person arrives and HR falls below 60 despite 30 s of effective ventilation, start compressions with oxygen if available.

5. At 4 minutes the HR is 130 and the baby starts to breathe and cry weakly. How do you manage afterwards?

> Stop ventilation once breathing well with normal HR. **Skin-to-skin** with the mother, cover; monitor breathing, HR, colour, temperature every 15 min for 2 h; early breastfeeding; check glucose if poor feeding. Record Apgars (e.g. 2, 5, 8), time ventilation started and duration. Watch for HIE (lethargy, poor suck, seizures): if present, **refer to SCBU**. Vitamin K, eye ointment, cord care. Counsel the mother.

Take-away: Prepare before every birth, dry and stimulate, and if the baby is not breathing, ventilate by 60 seconds. A rising heart rate tells you the ventilation is working.

Case 2: Advanced resuscitation at Mengo SCBU

At Mengo Hospital, an emergency caesarean is done for placental abruption at 37 weeks. The baby is born pale and floppy, with no respiratory effort. The neonatal team is present with oxygen, a pulse oximeter and emergency drugs.

1. What are the initial steps in the first minute?

> Radiant warmer, dry, remove wet towels, hat, position the head, clear airway if needed, brief stimulation, then **bag-and-mask ventilation by 60 s** in **21% oxygen**; attach the **pulse oximeter to the right hand**; helper assesses HR.

2. After 30 s of effective ventilation (chest rising), HR is 50. What do you do?

> Start **chest compressions** (two-thumb, lower third of sternum, one-third AP depth) coordinated **3:1** with ventilation (90 compressions + 30 breaths per min); increase oxygen to **100%**; consider **intubation** to secure the airway; reassess HR after 60 s.

3. After 60 s of compressions, HR is still 50. The baby is very pale with poor perfusion. What are the next steps?

> Place an **umbilical venous catheter**. Give **adrenaline 1:10,000, 0.1-0.3 mL/kg IV** (0.01-0.03 mg/kg), repeat every 3-5 min if HR remains below 60. Given **abruption and pallor**, suspect **hypovolaemia**: give **normal saline 10 mL/kg IV over 5-10 min** (O-negative blood if available and severe). Continue compressions and ventilation.

4. At 8 minutes HR rises to 120 and the baby gasps. What post-resuscitation care is needed?

> Admit to SCBU. Oxygen titrated to SpO2 targets (wean to 91-95%); normothermia and **avoid hyperthermia**; glucose check (treat below 2.6 mmol/L with 10% dextrose 2 mL/kg); **IV 10% dextrose 60 mL/kg/day**, restrict if oliguric; monitor for **seizures** (phenobarbital 20 mg/kg if they occur); FBC and blood culture; **ampicillin + gentamicin** if sepsis risk; monitor urine output; Sarnat grading; family counselling and documentation.

5. How will you counsel the mother and what follow-up is needed?

> Explain honestly what happened, that the baby needed advanced resuscitation, the risk of brain injury, and the plan. Involve the father/family. Arrange **developmental follow-up** (neurodevelopmental clinic, hearing and vision screening) and support breastfeeding (expressed milk).

> **Take-away:** Compressions only after 30 s of effective ventilation; adrenaline and volume come after that. Think hypovolaemia when a pale baby is born after APH.

Case 3: HBB skills station: troubleshooting ventilation

OSCE at UCU: a NeoNatalie simulator is 'born' not breathing. The candidate dries and stimulates, then starts ventilation at 50 seconds, but the examiner says 'the chest is not moving'.

1. What do you do first?

> **Reapply the mask**: chin first, then over mouth and nose, firm C-grip seal, lift the jaw into the mask. Check the mask is **size 1** and not covering the eyes.

2. The chest is still not moving. What next?

> **Reposition the head** into slight extension (sniffing position; shoulder roll if needed; not over-extended), then **clear the airway** (suction mouth then nose if secretions), **open the mouth slightly**, and **squeeze the bag harder**.

3. Now the chest rises. At what rate and how do you keep time?

> **40 breaths per minute**, saying 'Squeeze-two-three'. Continue while the helper assesses the HR after 1 minute.

4. The helper reports HR 120 but the baby is not yet breathing on its own. What do you do?

> HR is **normal**: **continue ventilation** until the baby breathes well spontaneously; then stop and monitor with the mother (breathing, HR, colour, temperature, feeding).

5. What must you do after the scenario?

> Communicate with the mother; **document** time of birth, time ventilation began, duration, HR response, Apgar; **decontaminate** the bag and mask (disassemble, wash, disinfect, dry, reassemble) and **test** it so it is ready for the next birth.

Take-away: When the chest does not rise, fix the mask seal and head position first. Ventilate at 40/min until the baby breathes, and always leave the equipment clean and tested.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.